



SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

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BITE497J – Project I

B.Tech (IT) Zeroth Review

Register Numbers	23BIT0279, 23BIT0221, 23BIT0251
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Project Code (Course Code)	BITE497J – Project I
Project Domain	Artificial Intelligence - Natural Language Processing and Neuro-Symbolic AI (applied to Governance / Public Policy Informatics)
Project Title	SchemeLogic - From Unstructured Scheme Documents to Verifiable Eligibility Logic: A Neuro-Symbolic Benchmark and Pipeline for Indian Welfare Schemes
Abstract (Mini-200 Words)	India operates more than 3,000 central and state welfare schemes , yet citizens routinely fail to discover benefits they qualify for, and LLM-based assistants that answer eligibility questions directly are prone to hallucinating criteria - a failure with real financial and social cost. While LLM-based extraction of executable rules from legal and regulatory text is an active research area , prior work has neither targeted the Indian welfare -scheme domain , nor examined how extraction errors distribute across predicate categories, nor measured whether extracted rule bases silently retain superseded criteria after scheme amendments. We propose SchemeLogic, a neuro-symbolic pipeline in which a large language model converts scheme documents (in English and Indian languages) into an auditable , executable eligibility -logic schema , while a deterministic symbolic engine - never the LLM - renders eligibility verdicts with full rule traces . We will release a gold -annotated benchmark of 100–150 Indian schemes with category -tagged eligibility predicates , and study (i) a category -wise failure taxonomy of extraction errors, (ii) cross-lingual extraction consistency between Indic -language and English notifications , (iii) temporal robustness of rule maintenance under scheme amendments , and (iv) the effectiveness of confidence-based human deferral, using harm-

	weighted metrics that separate false-positive from false-negative eligibility verdicts.
Keywords	Large Language Models, Retrieval-Augmented Generation, Neuro-Symbolic AI, Eligibility Extraction, Welfare Schemes, Rules-as-Code, Cross-Lingual NLP
Approval Status	YES
Meeting date & Time	17/07/2026
Student Guide Interaction meeting	The points discussed during the meeting Discussed the Literature and finalized the title and outcomes
Guide Name	Dr. Vivekananda GN
Guide Signature	
Approval Date	17/07/2026

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